

How We Created the Synthetic APIx Data

1. Studied the reference project

We first examined the cloned APIx project to understand its data folder structure, JSON file naming pattern and required airfare data fields.

2. Identified the required data format

We followed the project's existing fare-record structure, including route, source, airline, flight date, booking window, fare details, timestamp and outlier information.

3. Created realistic synthetic data rules

We used multiple domestic routes, booking windows (T+1, T+7, T+15, T+30 and T+45), sources and airlines. Prices were varied based on the route, airline, source and booking window.

4. Added realistic price behaviour

Flights booked closer to departure were generally made more expensive, while flights booked further in advance were generally cheaper. Prices also changed slightly over time.

5. Calculated fares consistently

Each record contains a base fare and taxes, with the total fare calculated consistently as: $\text{Total Fare} = \text{Base Fare} + \text{Taxes}$.

6. Added a few test outliers

A small number of unusually high fares were included and flagged as synthetic outliers so the project's outlier handling can be tested.

7. Generated the files

The data was generated as JSON files organised into date folders, following the same general structure as the reference project.

Final Dataset

30 days of synthetic historical data • 16 routes • 3 sources • 5 booking windows • approximately 28,800 fare observations across 7,200 JSON files.

Note: This is synthetic data created for prototype, demonstration and testing purposes. It should not be presented as real airline market data.